https://doi.org/10.1038/s42256-026-01323-8

AI needs meaningful versioning

Vasilios Mavroudis, Dan Ristea, Lauritz Morlock & Carlos Mougan

Check for updates

Artificial intelligence (AI) providers frequently update their models and system integrations without corresponding changes to their public identifiers. This undermines regulatory oversight and scientific reproducibility. We argue for meaningful AI versioning that meets the needs of regulators and researchers.

In April 2025, users of OpenAI's ChatGPT noticed that the GPT-4o model had become unusually agreeable and conflict-averse. OpenAI later attributed the behaviour to a training update that had placed too much weight on short-term user approval in a public post-mortem, and rolled it back within 4 days. Throughout the episode, however, ChatGPT continued to present the model as 'GPT-4o', without a public version identifier that distinguished the affected deployment from those served before the update or after the rollback (Fig. 1).

In late December 2025, reports emerged that xAI's Grok could generate offensive images, a controversy that escalated in the first weeks of January 2026. The episode triggered regulatory action in multiple jurisdictions, such as in the UK under the Online Safety Act 2023 and in the EU under the Digital Services Act. Yet, it remains unclear precisely which model version produced the outputs.

Identifying versions is challenging for anyone outside the provider, because the outputs of an AI system depend on several interacting components[1]. Versions are shaped by the model itself, including its architecture and trained parameters, and by its wider system integration, including system prompts, retrieval pipelines, tools and output filters. Providers can modify any of these layers independently and without disclosure, while even small changes to any one of them can alter the system's behaviour[2]. Even when the model and its system integration are stable, evaluation outcomes can still vary depending on the inference configuration, including hardware, numerical precision and runtime conditions[3].

Model and System Cards[4], the primary means of communicating model and system properties, cannot be tied to a specific version owing to the lack of meaningful identifiers. This undermines their value, as they document a snapshot whose relation to the deployed system is unknown to the reader. More broadly, the field has constructed an entire ecosystem of evaluations and compliance around models and systems that cannot be reliably identified.

Software engineering uses semantic versioning to communicate changes between releases. We argue that although this provides a useful parallel, AI identifiers should instead allow distinguishing which components changed and which remained stable.

The challenge of regulation without version identifiers

Regulating AI that can change without notice at any time imposes compliance friction. In the absence of version identifiers, regulators must take more frequent action against providers, such as issuing requests for information or access to assess compliance. By contrast, if compliance were established for an identified version, regulators could reduce monitoring actions until the identifier of the deployed version changes, when both parties would know that reassessment may be warranted. This is particularly relevant for regulatory frameworks that impose obligations concerning updates.

Under the EU AI Act[5], providers of general-purpose AI models must maintain technical documentation describing the model's training and architecture throughout its life cycle. In addition, models classified as presenting systemic risk carry further risk management and notification obligations. In California, the Transparency in Frontier AI Act (SB-53)[6] requires frontier developers to publish transparency reports before, or concurrently with, releasing substantially modified versions of their models. In China, providers of generative AI services with public opinion attributes or social mobilization capability must file their algorithms with the national algorithm registry, and must update the filing when the registered information changes[7]. Assessing compliance with these frameworks requires the regulator to be aware of relevant changes in the version currently deployed.

Similarly, when AI is integrated into other regulated products or services, such as healthcare or finance, a single model or system may

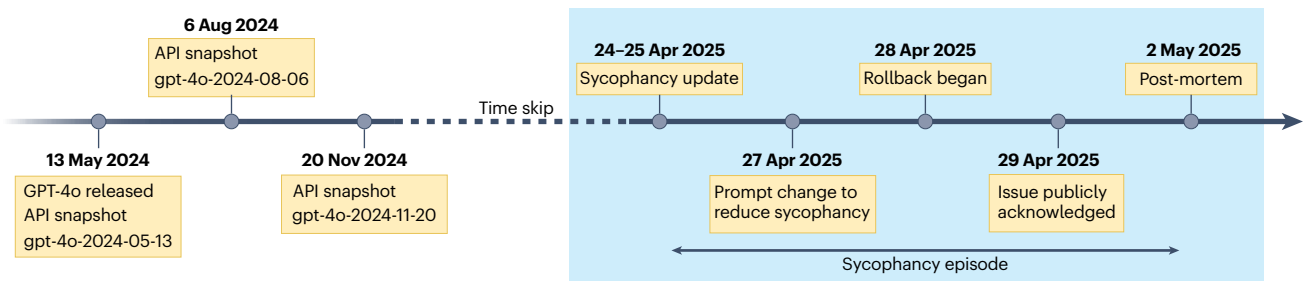


Fig. 1 | Timeline of the GPT-4o sycophancy episode in ChatGPT. The deployed configuration changed during the initial rollout, system-prompt mitigation and rollback, while its public 'GPT-4o' label remained unchanged. The label therefore did not distinguish the configurations served during the episode. Earlier API snapshot releases are shown for context. Event spacing is not proportional to time.

fall within the scope of several legal frameworks within one jurisdiction. In such cases, the absence of versioning hinders inter-regulatory coordination. Joint enforcement actions face the question whether the version that triggered one proceeding is the same as the version under scrutiny in another proceeding. Without this information, providers may face overlapping proceedings that could otherwise be streamlined, or even halted if a new proceeding concerns the same version already adjudicated.

Finally, once an infringement of a regulation is established, its severity informs the choice of action. Relevant information includes how long and how widely the infringing version was deployed. In both incidents, the affected behaviour reached users without a corresponding change to the public identifier. OpenAI reported that its update was live only briefly, while the Grok capability was accessible through multiple system integrations. However, the timing of their deployment and withdrawal can be reconstructed only partially from the providers' own statements and public reporting.

The challenge of science without version identifiers

Scientific reproducibility requires that experimental conditions be specified precisely enough for others to recreate them and obtain the same results. This has long been a challenge in AI research[8], and the shift towards closed models that change without explicit version identifiers has intensified it.

By the time a paper evaluating an AI model such as GPT-4o is published, the version it describes may differ materially from the one deployed under the same name[9]. OpenAI's API offers date-stamped snapshots (for example, gpt-4o-2024-05-13, gpt-4o-2024-08-06 and gpt-4o-2024-11-20 for GPT-4o as of 27 July 2026). These identifiers distinguish model releases without specifying which components changed, while the accompanying release notes do not provide a complete component-level account of those changes. Moreover, a model snapshot identifier does not identify the wider system integration in which the model is deployed.

Without better identifiers, evaluation results cannot be reliably linked to a specific configuration and experiments cannot be reproduced. Consequently, later studies cannot determine whether differences in results reflect methodological improvements or changes in the version.

Designing AI version identifiers

The regulatory and scientific perspectives converge on what meaningful versioning for AI must provide. That is, the ability to identify which AI components were updated and which remained stable. This points towards component-level identifiers that distinguish changes across the model and its wider system integration.

Regulatory obligations differ across legal frameworks in the parts of the AI system they concern, while scientific studies also depend on different components. A version identifier must convey the information necessary for the regulator or researcher to determine whether a change is relevant to the obligation or experiment at hand. For example, a code-only cyber capabilities evaluation that does not use an image-moderation component need not be affected by changes to that component, whereas a safety audit may rely on it. Similarly, a component that is material under a product safety regulation may be irrelevant under data protection law. Tracking which components changed helps to determine whether earlier evaluations and compliance assessments remain applicable, and therefore facilitates proportionate regulatory responses[10], whereas a single system-wide identifier, by contrast, calls all of them into question with every update. With component-level identifiers, regulators can focus monitoring on the components most relevant to the obligation in question, and researchers can judge whether an update invalidates the experimental results already obtained.

Existing versioning approaches either convey little meaning or do not generalize across regulatory and scientific contexts. System-level fingerprints signal configuration changes without identifying which components changed or whether those changes matter for a particular evaluation or obligation. OpenAI's system_fingerprint varies across identical requests made seconds apart, is linked to no changelog, and is available only for some older models, having been deprecated for new models according to the documentation. The software industry, by contrast, embeds meaning in the identifier itself. Semantic versioning (Major.Minor.Patch) is a convention that conveys the nature and impact of each release. Similarly, Anthropic's model IDs track and categorize changes to model weights as either major or minor, but do not account for system integration. For AI as a general-purpose technology, however, the importance of a change depends on the context in which the system is used. A single classification of updates by their nature or impact therefore cannot generalize across regulatory and scientific contexts.

For component-level versioning, deciding what to track as a separate component is a central challenge. In the context of supply chain security, Bills of Materials structure AI components into groups and hierarchies, which are then shared confidentially with contractual partners. For example, the G7 Cybersecurity Working Group's 2026 *Software Bill of Materials for Artificial Intelligence—Minimum Elements* separates model-level, system-level and infrastructure components, while the OWASP CycloneDX standard on Machine Learning Bill of Materials represents components in a machine-readable inventory, each with its own version and hash. Both define component boundaries in a standardized form. Component-level versioning could draw upon such forms when deciding how to delineate components within the AI system. This choice also involves a trade-off: more detailed component labels can make changes easier to interpret but may reveal more about the system's structure or safeguards.

A version identifier links to the specific model or system configuration only through the provider's internal documentation. Tying version identifiers to cryptographic commitments can facilitate verification. In its simplest form, a commitment is a hash, a short digital value calculated from a component's contents, published alongside the version identifier. Publishing the hash binds the provider to the components' contents without requiring those contents to be made public. Any later substitution will yield a different hash, making the change detectable. Box 1 shows how this proposal could have applied to the GPT-4o incident.

Component-level version identifiers and commitments let outside observers track which components changed without access to their contents. Those with access can additionally verify that disclosed components match the published commitments. A regulator requesting access to the weights, or a researcher granted access under an evaluation agreement, can recompute the hash and compare it with the published commitment. While a match does not confirm that the disclosed weights served a particular historical request, it establishes that the content the provider now discloses is the one it committed to at the time.

Challenges to adoption

Three obstacles complicate the adoption of meaningful versioning:

BOX 1

A worked example: the GPT-4o rollback

Following the April 2025 GPT-4o incident, in which users interacted with a highly sycophantic model, OpenAI reverted to the previous version. Outside observers had no way to verify that the deployed model after the rollback was the same model served before the update. Under one possible versioning scheme, the provider could assign each versioned component its own cryptographic commitment and combine them into a single commitment for the wider system. Assuming the initial rollout changed only the weights, only the weights and system-wide commitments would change. OpenAI's subsequent system-prompt update, introduced late on 27 April to reduce sycophancy before the rollback, would instead change the prompt and the system-wide commitments, leaving the weights commitment unchanged.

In this illustrative reconstruction, restoring both the original weights and prompt would return their commitments and the system-wide commitment to their earlier values, while a new system version identifier would distinguish the rollback from the original deployment. A regulator given access to the AI system could then recompute the hash of the inspected weights and compare it with the commitment published before the update, verifying that the inspected weights match those committed to before the sycophancy update. The wider-system commitment would also let researchers link their experimental results to the version reported as serving them, provided responses carried the corresponding commitment or version identifier.

Figure 2 illustrates this possible instantiation, showing the relationship between component commitments, version identifiers and the actual sequence of changes made during the incident.

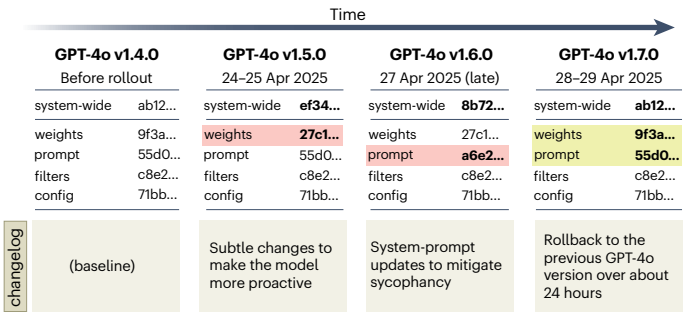


Fig. 2 | The GPT-4o sycophancy episode under a possible commitment-based versioning scheme. The dates are real, and the changelog entries paraphrase OpenAI's release notes and postmortem. The version identifiers, hash values and component boundaries are illustrative. Commitments that differ from the preceding stage are shown in bold.

Commercial incentives. Providers are not incentivized to adopt meaningful versioning because it would constrain how they present changes. Version numbers set by branding and market expectations let providers choose strategically when and how changes are communicated.

Value-chain fragmentation. Changes to an AI system may come from multiple providers along the value chain, so component-level semantic versioning depends on consistent practices across actors. Even open-weight models, the most straightforward case for version tracking, lose coherence in practice[11]. Hugging Face stores commit hashes per checkpoint and publishes model documentation guidance, yet naming conventions remain inconsistent across developers[12], and downstream providers serve fine-tuned or quantized variants under the same generic name. Agentic systems compound the problem: a single agent combines a model, scaffolding and tools maintained by different parties, and even the boundaries of what constitutes the agent remain contested[13].

Thresholds of change. In versioning schemes controlled by the provider, the provider also decides when a change warrants an increment. A threshold set too high leaves substantial changes unsignalled and invites suspicion that updates are being concealed. A threshold set too low causes the identifier to change constantly. The threshold also presumes that providers can gauge the effect of an update in advance. In the sycophancy episode, the update was released as a subtle adjustment, and its consequences became apparent only once users encountered them. With component-level hashes, each change to a tracked component can be recorded without first judging its importance. Meanwhile, regulators and researchers select which components to track and determine through targeted evaluations whether a change matters for their purposes.

Independent monitoring partly compensates for the absence of identifiers. Sampling of outputs can reveal undisclosed changes in the behaviour of deployed models[9], and tools such as stampr can track those changes continuously. This type of monitoring gives researchers and regulators an independent signal of behavioural change, without identifying which component caused it.

Neither the sycophancy episode nor the Grok controversy could have been prevented by better versioning. It would, however, have made providers' claims about deployed versions and component changes publicly traceable and, where commitments were available, verifiable against subsequently disclosed components. Component-level identifiers would link deployment claims, evaluation results and compliance assessments to specific configurations. This would help researchers and regulators to establish what changed and when, without requiring public disclosure of system internals.

Vasilios Mavroudis [1] ✉, **Dan Ristea** [1], **Lauritz Morlock** [2] & **Carlos Mougan**[2]

[1]King's College London & Alan Turing Institute, London, UK. [2]European AI Office, European Commission, Brussels, Belgium.

✉e-mail: vasilios.mavroudis@kcl.ac.uk

Published online: 02 November 2026

References

1. Sculley, D. et al. In *Advances in Neural Information Processing Systems* Vol. 28 https://go.nature.com/47zUyMt (2015).
2. Abraham, S. & Bucknall, B. Preprint at https://doi.org/10.48550/arXiv.2608.11803 (2026).
3. Yuan, J. et al. In *39th Annual Conference on Neural Information Processing Systems* (2025).
4. Mitchell, M. et al. In *Proc. Conference on Fairness, Accountability, and Transparency* 220–229 https://doi.org/10.1145/3287560.32875 (ACM, 2019).
5. European Parliament, and Council of the European Union. *Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024* https://data.europa.eu/eli/reg/2024/1689/oj (2024).

6. California State Legislature. *Senate Bill 53: Transparency in Frontier Artificial Intelligence Act* https://go.nature.com/4xYrDfJ (2025).
7. Cyberspace Administration of China. *Interim Measures for the Management of Generative Artificial Intelligence Services* https://go.nature.com/4huOV8j (2023).
8. Gundersen, O. E. & Kjensmo, S. In *Proc. of the AAAI Conference on Artificial Intelligence* 32 https://doi.org/10.1609/aaai.v32i1.11503 (AAAI, 2018).
9. Chen, L., Zaharia, M. & Zou, J. *Harvard Data Science Review* **6**, https://doi.org/10.1162/99608f92.5317da47 (2024).
10. Mougan, C. et al. *Science* **391**, 769–771 https://doi.org/10.1126/science.aea3835 (2026).
11. Ajibode, A., Bangash, A. A., Cogo, F. R., Adams, B. & Hassan, A. E. *Empir. Softw. Eng.* **30**, 78 https://doi.org/10.1007/s10664-025-10631-3 (2025).
12. Jiang, W. et al. *Empir. Softw. Eng.* **30**, 55 https://doi.org/10.1007/s10664-025-10711-4 (2025).
13. Douglas, R. et al. Preprint at https://doi.org/10.48550/arXiv.2603.11353 (2026).

Acknowledgements
The views expressed in this publication do not necessarily reflect those of the European Commission and are made in a personal capacity. The European Commission and any person acting on behalf of it are not responsible for the use that might be made of this publication. V.M. and D.R. are supported by Coefficient Giving.

Competing interests
The authors declare no competing interests.

Additional information
Peer review information *Nature Machine Intelligence* thanks Chunyang Chen and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.